

KopiMod

Alat pengambilan keputusan bagi petani kecil Jawa Barat.

User Manual



1 Overview of KopiMod

KopiMod (Kopi Model or Coffee model) is an excel based decision-making tool and has been developed as part of the *Agricultural Policy Research to Support Natural Resource Management in Indonesia's Upland Landscapes* project, funded by the Australian Centre for International Agricultural Research (ACIAR project ADP/2015/043).

The objective of KopiMod is to provide a flexible, user-friendly tool to allow farmers, extension officers and other working in Agriculture in West Java make decisions about land use and farm management decisions. The tool seeks to present revenue, costs and labour requirements across a number of different crop types to understand the trade-offs of different decisions.

KopiMod has used data collected from 499 households on the slopes surrounding Bandung, West Java (ACIAR Survey ADP043 BANDUNG 2019).

2 How to use – *Dasbor*

The Dasbor (dashboard) allows the user of the model to compare revenue, cost, and labour requirements of different land use options for smallholder farmers in West Java. This is achieved by estimating the revenue, costs and labour for the current land use of the plot, and then comparing this with an alternative land use scenario.

The Dasbor allows the user to select from four main land use types:

- Plots that primarily grow and harvest coffee.
- Plots that primarily grow and harvest rice.
- Plots that primarily grow and harvest annual horticultural products (including vegetables and tubers).
- Plots that primarily grow and harvest other agroforestry products such as timber and fruits.

The first step when opening the workbook is to accept the macros and any other permissions. These are required for the model to run correctly.

The Dasbor is designed to be stepped through one detail at a time. As details on plots are entered, new cells will appear for additional information on the current or alternative land use scenario.

1. Enter the general plot information
2. Enter the current land use of the plot, entering information when new options appear on the Dasbor.
3. Review the results on revenue and costs. If any assumptions need to change, return to the Second step.
4. Enter the alternative land use option scenario based on the land uses you want to compare, entering information when new options appear on the Dasbor.
5. Review the results on revenue and costs for the alternative land use.
6. Use the comparison module at the end of the module to compare revenues, costs and labour requirements of your current land use and the alternative land use scenario.

The information is presented on a plot basis, rather than on a household level. Smallholder farmers with multiple plots will need to save a new version of the model for each plot they want to compare land use options.

Note:

The agroforestry option also allows for the user to enter multiple crops such as coffee and horticulture. This land use option will be the most flexible for mixed gardens with coffee, horticulture, and agroforestry.

2.1 General plot information (Informasi plot umum)

The user is first required to enter general plot information for the plot of interest, including

- The name of the plot (for record keeping)
- The area of the plot (in hectares)
- The plot slope (flat sloping is assumed to be less than 15% slope, moderate between 15-30%, and steep slopes >45%).

The user can also enter the daily wage rate, the nominal discount rate, and the inflation rate. The daily wage rate can be either what a farmer would otherwise earn if not working on their plot, or what hired labour would cost. The discount rate can be set to the cost of loans they have access to. Otherwise, these values can be left to the default options of 60,000 rph per labour day, 15% discount rate, and 3% inflation rate.

Note:

Any cell that is highlighted in blue can have data entered in by the user. The remaining cells should be left unedited for the model to work correctly.



2.2 Current land use information (Informasi penggunaan lahan saat ini)

This module allows the user to enter their current land use information for the plot of interest.

Drop down menus are available for the user to select from a subset of available options. The first of these is the land use option, where the user can select from coffee, horticulture, rice, agroforestry, cleared land, or uncleared forest. The user should select the option that best described their property.

Figure 1 – The user is first required to enter the land use that best describes the land use on the plot currently.

Apa guna lahan tersebut saat ini?

Kategori

- Kopi
- Hortikultura
- Beras
- Agroforestri
- Lahan yang dibersihkan
- Hutan

After a selection is made from this drop-down list, relevant cells will appear below the drop down list in which the user can enter additional information. These are discussed for each land use option below:

2.2.1 Coffee (Kopi)

If the user selects coffee as their land use option, the following cells will appear for the user to enter additional information. These inputs are optional – if no number is entered into these cells the model assumes the median values within the survey data. But each piece of information will calibrate the model to better reflect the inputs and outputs of their plot.

- **The number of coffee trees on the plot.**
- **The percentage of these trees that are productive.** The survey data indicates very few plots have 100% productive plots, with flat/moderate coffee plots having only 65% of their trees productive, and steep slopes only 58%.
- **The kilograms of cherries harvested in the last year** – This will calibrate the output per tree in the model to match that experienced by the farmer. Otherwise a median value based on the slope of the plot will be assumed.
- **Average age of coffee trees** – The age of the coffee trees will have implications on coffee yields and when the trees need replacing. The model will assume an age of 15 years if no value is entered.
- **Do you only pick the ripe/red cherries? (Yes/No)** – Picking ripe red cherries can allow the farmer to receive price premiums for the coffee in certain value chains but requires additional labour.
- **Average price per kg of coffee cherries** – The price received for the coffee cherries

2.2.2 Horticulture (hortikultura)

If the user selects horticulture as their land use option, the following cells will appear for the user to enter additional information. These inputs are optional – if no number is entered into these cells the model assumes the median values within the survey data. But each piece of information will calibrate the model to better reflect the inputs and outputs of their plot.

- **How long have you been growing horticulture on this land?**
- **What was the total income received from crops across all seasons last year – including the value of any crops consumed within the household.**
– Horticultural crops in the survey data have multiple seasons, and often much of the produce is consumed within the household. This value seeks to combine the income of all crops harvested across the seasons, and the value of the production consumed within the household. A median value from the survey is assumed if this is left blank.

2.2.3 Paddy (beras)

If the user selects horticulture as their land use option, the following cells will appear for the user to enter additional information. These inputs are optional – if no number is entered into these cells the model assumes the median values within the survey

data. But each piece of information will calibrate the model to better reflect the inputs and outputs of their plot.

- **How long have you been growing rice on this land?**
- **Price per kg of rice**
- **How much Rice did you produce last year in all seasons - including what you sell or consume in your household? –** Paddy crops in the survey data have multiple seasons, and often much of the produce is consumed within the household. This value seeks to combine the production of all paddy crops harvested across the seasons, and the value of the production consumed within the household. A median value from the survey is assumed if this is left blank.

2.2.4 Agroforestry (agroforestri)

Mixed agroforestry systems often have a several crops grown in combination. If agroforestry land use is chosen, the user is able to select the following crops grown on the plot.

- Coffee (Kopi)
- Horticulture (horticulture)
- Fruit tree products (buah pohon)
- Cinnamon (kayu manis)
- Timber trees (kayu pohon)

The following questions will appear based on the selection of crops on the plot:

- For coffee, the same questions as in the coffee module are asked.
- No questions are asked about horticulture, with median costs and revenue assumed from horticultural products.
- For fruit trees:
 - The number of fruit trees
 - The percentage of these trees that are productive
 - The average age of fruit trees
- For cinnamon trees:
 - The number of cinnamon trees.
 - The average age of the cinnamon trees.
 - The average price received for wet bark on cinnamon trees.
- For timber trees:
 - The number of timber trees
 - The average age of the timber trees
 - The average price received per m³ of timber harvested.

Note

For the coffee, horticulture, and rice modules the additional data entered is optional. For the agroforestry module, the additional data is required, in particular the number of coffee, cinnamon, fruit and timber trees on the plot. The number of trees is used to estimate the proportion of the plot dedicated to each commodity type. If no values are entered for the number of trees, it is assumed the entire plot is used for horticulture.

2.2.5 Cleared land or forest (Lahan yang dibersihkan/Hutan)

For cleared land or forest, no additional information is required from the user. It is assumed no revenue is derived from the plot and requires no labour or costs.

The user may choose these options as they might be considering their options for converting the land for production. The model allows for the comparison of options in the later alternative land use option module.

The distinction between cleared land or forest matters when considering the labour required for converting the plot to different land uses.

Note

To clear all existing data for the current land use, the user can press the ***Hapus semua informasi penggunaan lahan saat ini*** button.

2.3 Results - current land use (Pendapatan dan biaya - Penggunaan lahan saat ini)

After information is entered for the current land use, the results are ready to review in the following module.

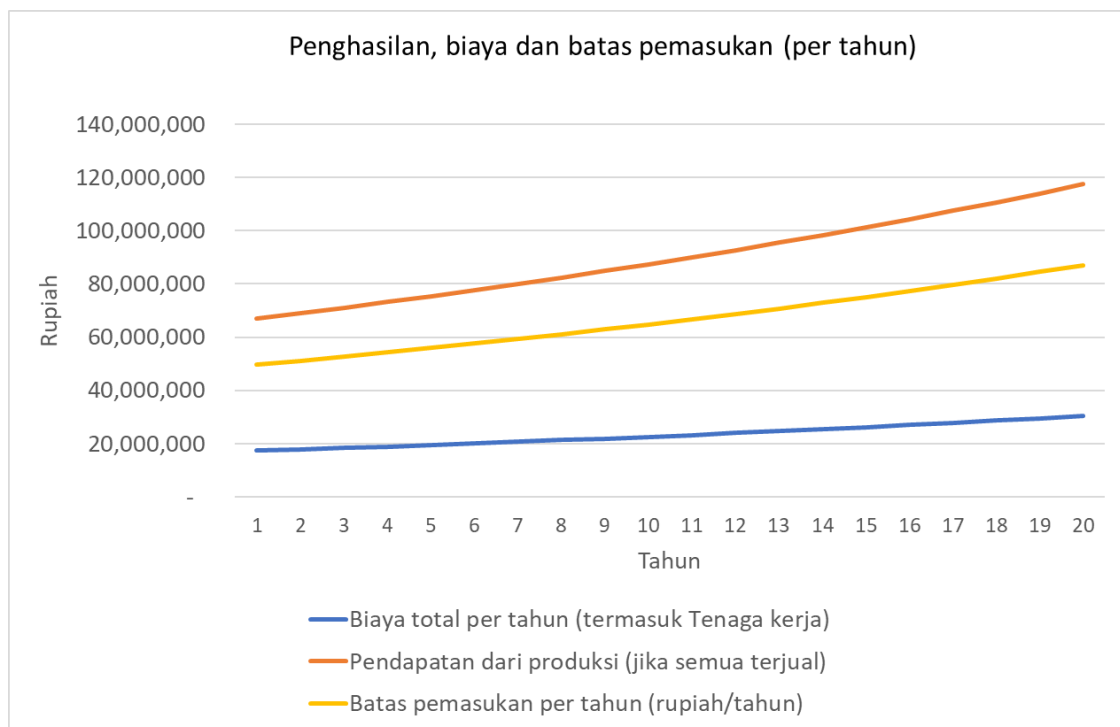
Revenues, costs, and labour requirements are estimated over a 20-year period. More detail on each model output is in the table below:

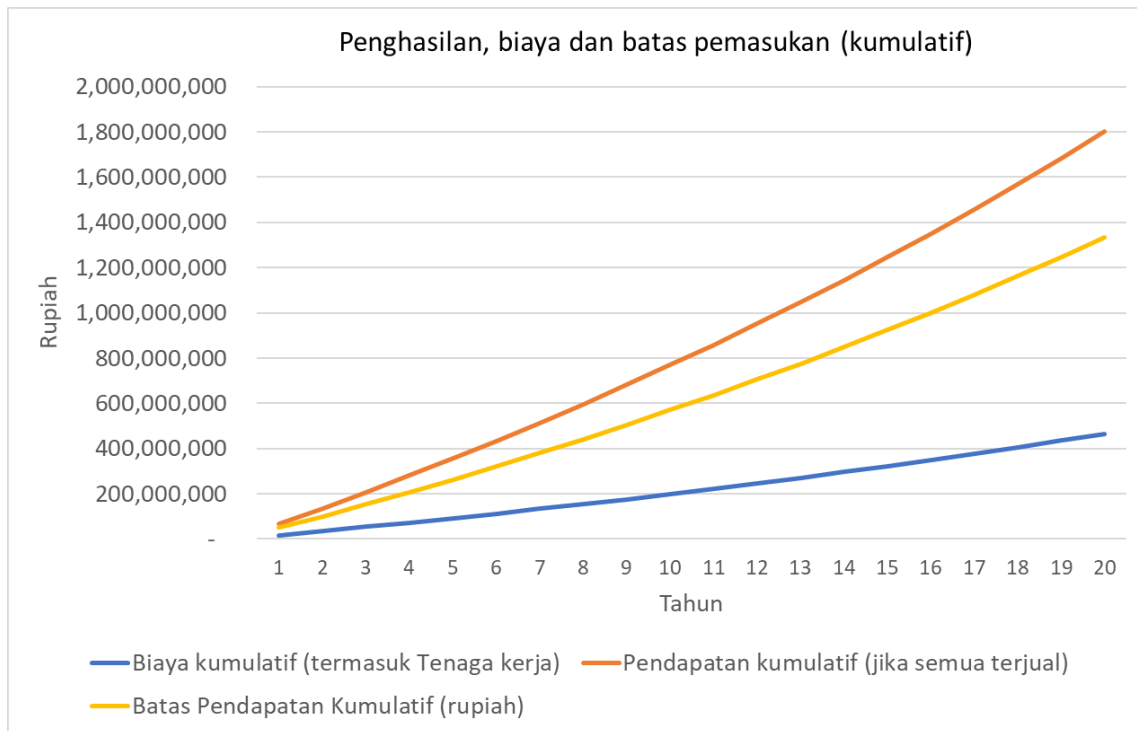
Model output	Description
Seed costs	Total costs for seeds or seedlings each year – in rupiah.
Fertiliser costs	Total costs for fertilisers, mulch or related applications each year – in rupiah.
Chemical costs	Total costs for pesticides and herbicides each year – in rupiah.

Total costs	Total of above costs
Number of labour days required per year	The number of family or hired labour days required for the land use.
Labour costs	The number of family or hired labour days required for the land use, multiplied by the daily wage rate.
Total costs	Total costs including labour.
Total costs – cumulative	The cumulative total costs of the land use. So by year 20 this value will equal the total costs for the land use option over the previous 20 years.
Income from production	Income from harvests from the plot.
Income from production – cumulative	The cumulative income from harvests from the plot. So by year 20 this value will equal the total income for the land use option over the previous 20 years.
Net income per year	Total income from production, minus total costs.
Net income– cumulative	The cumulative net income. So by year 20 this value will equal the net income for the land use option over the previous 20 years.
Net income a year - discounted	The net income, discounted at the nominal discount rate provided in the Dasbor.
Discounted cumulative cash flow	The cumulative net income, discounted at the nominal discount rate provided in the Dasbor.
Net present value	The sum of the net income a year, discounted, representing the net present value of the land use over 20 years.
Value of timber trees	For plots with timber, it is assumed these trees remain standing. This value gives the value of these trees (equal to the revenue received if harvested in that year).
Marginal increase in value of timber trees	The change in the value of timber trees from the year prior. Represents the additional income that could be received if the trees are left standing for another year.
Total net income a year, including growth in the value of timber trees.	Total net income a year, including the marginal increase in value of timber trees.

Net income– cumulative – with value of timber trees	The cumulative net income a year, including the marginal increase in value of timber trees.
Net income a year - discounted	The net income, discounted at the nominal discount rate provided in the Dasbor.
Discounted cumulative cash flow	The cumulative net income, discounted at the nominal discount rate provided in the Dasbor.
Net present value	The sum of the net income a year, discounted, representing the net present value of the land use over 20 years.

The information is also presented in the charts below. The first chart plots revenue, costs and net income per year. The second plot shows the cumulative revenue, costs and net income a year.





2.4 Alternative land use information (Informasi penggunaan lahan saat ini)

The alternative land use information module now allows the user to compare revenue, costs and labour requirements for alternative land use options. This allows for smallholder farmers to understand the trade-offs associated with changing land uses.

The model also allows for smallholder farmers to understand the trade-offs associated with different management practices. These scenarios include:

- The changes in revenue due to changes in price of coffee, rice, cinnamon, or timber.
- The costs and revenue from harvesting only ripe, red cherries.
- The costs and revenue from stumping/replanting coffee trees.
- The costs and revenue from replanting timber or fruit trees.

2.4.1 Testing different land uses

Most scenarios will involve a change in land use type (e.g. from coffee to horticulture, horticulture to rice etc).

As with the current land use information module, the user is first asked what land use they want to consider. This can be achieved with the drop-down menu as shown below.

Penggunaan lahan baru apa di plot ini yang ingin Anda pertimbangkan?

Kategori

Kopi
Hortikultura
Beras
Agroforestri

With this selection, new options will appear for additional information. For coffee, horticulture and rice these options include inputting the price of outputs.

For agroforestry a few additional inputs are required. The user is first asked for the percentage of the plot to be used for coffee, horticulture, fruits, cinnamon, or timber as shown below.

Berapa persentase plot yang Anda inginkan untuk memiliki Pohon kopi?		%
Berapa persentase plot yang Anda inginkan untuk memiliki Buah pohon?		%
Berapa persentase plot yang Anda inginkan untuk memiliki Kayu manis pohon?		%
Berapa persentase plot yang Anda inginkan untuk memiliki Kayu pohon?		%
Berapa persentase plot yang ingin Anda tanami Hortikultura?		%

As the area of the plot used for each commodity is entered, new cells will appear for additional information. These relate to prices to outputs, but in some instances relate to whether new trees or planted or old ones retained. These options will only appear if coffee or agroforestry is the current land use, and these cells are in sections 2.4.4 and 2.4.5.

2.4.2 Testing the trade-offs from changes in the output price.

Smallholder farmers may want to see what happens if prices for outputs change. For example, what would revenue look like if the price of rice, coffee or timber changes.

To see this scenario, in the “alternative land use scenario”, pick the alternative land use to match the current land use of the plot (e.g. if rice is currently harvested on the plot, pick rice for the alternative scenario). Here, an option will appear to enter a new price for the crop as shown below.

Beras

Harga per kg untuk Beras untuk skenario baru

Rupiah/kg

This scenario can be tested for coffee, rice, fruits, timber and cinnamon.

Unfortunately the option is not available for horticulture because of the variety of crops grown in these plots.

2.4.3 Testing the trade-offs from harvesting only ripe, red coffee cherries

In some coffee value chains, the harvesting of ripe, red coffee cherries can result in higher prices for the coffee as the beans tend to be of higher value. However, this can also increase harvest labour requirements.

Smallholder farmers may want to understand the trade offs associated with harvesting only red cherries. To achieve this, enter coffee as the alternative land use option (assuming coffee is also the current land use and that they do not currently harvest only red cherries), and select “yes” for only harvesting red coffee cherries.

Just doing this will only change the labour requirements. If price premiums can be made from this harvesting approach, enter a new price per kg of coffee cherries. Doing this will also change the revenue.

Untuk skenario ini, apakah Anda hanya akan memilih ceri merah yang matang?	Ya	
Harga per kg kopi cherry untuk skenario ini	9000	Rupiah/kg

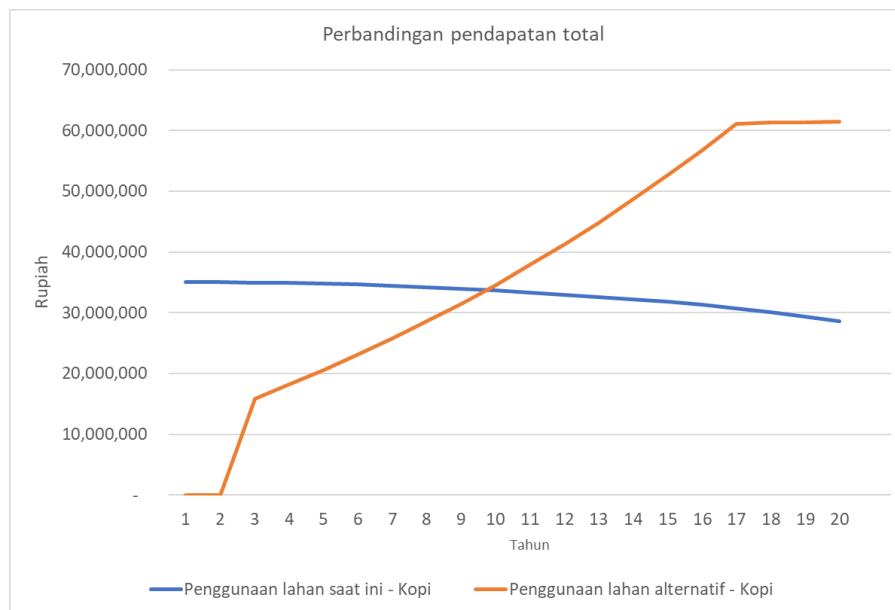
2.4.4 Testing the trade-offs from stumping/replanting coffee trees

Old coffee trees will be less productive than younger coffee trees. Stumping or replanting coffee trees is a common strategy for increasing the yield of coffee trees in the medium/long term. However, this approach also has costs associated with stumping, and foregone revenue.

Understanding these trade-offs is straightforward in the model. First the user should select coffee as the alternative land use, and then select yes for replanting or stumping the coffee trees.

Penggunaan lahan baru apa di plot ini yang ingin Anda pertimbangkan?	Kopi	kategori
Kopi Untuk skenario ini, apakah Anda ingin menguji apa yang akan terjadi jika Anda tunggu atau mengganti semua Pohon Kopi Anda?	Ya	

The model assumes a cost associated with stumping the trees, and the time it takes for harvests to return. An example of the results is shown below, with revenue initially at zero for the first years but increases to be much higher after 10 years compared to a plot starting with older coffee trees.



Note

The same scenario can also be entered if:

- The current land use is coffee, and the alternative land use introduces other trees and crop in an agroforestry model.
- The current land use is agroforestry, and the alternative land use scenario is changing the proportion of crops (but still mostly a mixed agroforestry system)
- Moving from agroforestry with coffee, to entirely coffee.

Under both scenarios, the prompts will appear in the alternative land use scenario module when relevant. Stumping and replanting will only occur on trees that remain under each scenario.

2.4.5 Testing the trade-offs from replanting fruit and/or timber trees

As with coffee trees, stumping or replanting fruit or timber trees can improve yields if the trees are old, but comes at a cost initially due to upfront land preparation costs and foregone revenue. With timber however, there is some upfront revenue from the sale of timber.

For this scenario, the user should select agroforestry as the alternative land use scenario (assuming it is also the current land use with timber and/or fruit trees).

Once this is selected, a prompt will appear asking whether they would like to replant any existing trees? If yes, new prompts will appear asking if they want to replant any fruit or timber trees.

Apakah Anda ingin menguji Pendapatan dan biaya jika Anda menebang atau menanam kembali salah satu pohon Anda?	Ya	<i>kategori</i>
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Pakah Anda ingin menanam kembali Buah pohon Anda (jika ada)? Apakah Anda ingin Memanen dan menanam kembali Kayu pohon Anda (jika ada)?	Ya
	Ya

The model assumes a cost associated with stumping the trees, and the time it takes for harvests to return. It also assumes that the timber is sold in the first year when the trees are replanted - increasing the revenue in the first year.

2.5 Results - Hasil (Penggunaan lahan alternatif)

The results for the alternative land use scenario are presented the same as for the current land use scenario. Revenue, costs and labour requirements are all profiled over 20 years.

2.6 Comparison of scenarios (Perbandingan penggunaan lahan)

The final module on the Dasbor is a comparison between the current and alternative land use options. This is where the user can see the difference in revenues, costs and labour requirements of different scenarios side by side.

The information presented on this sheet includes:

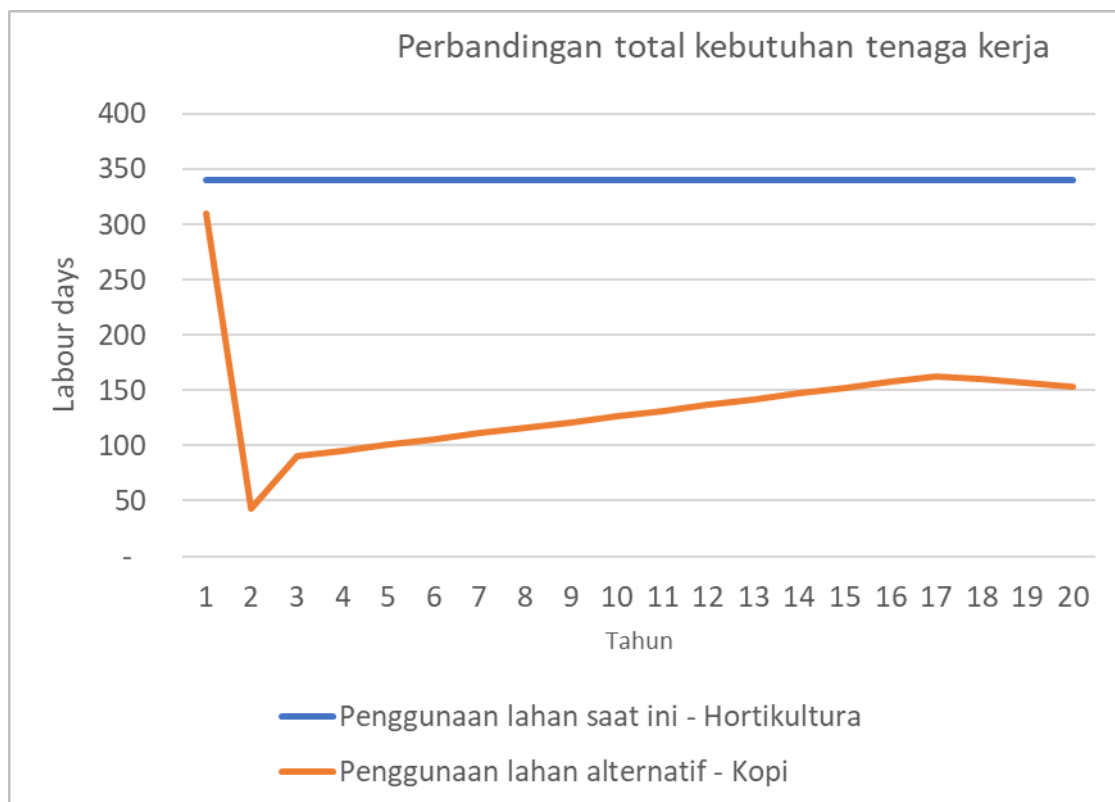
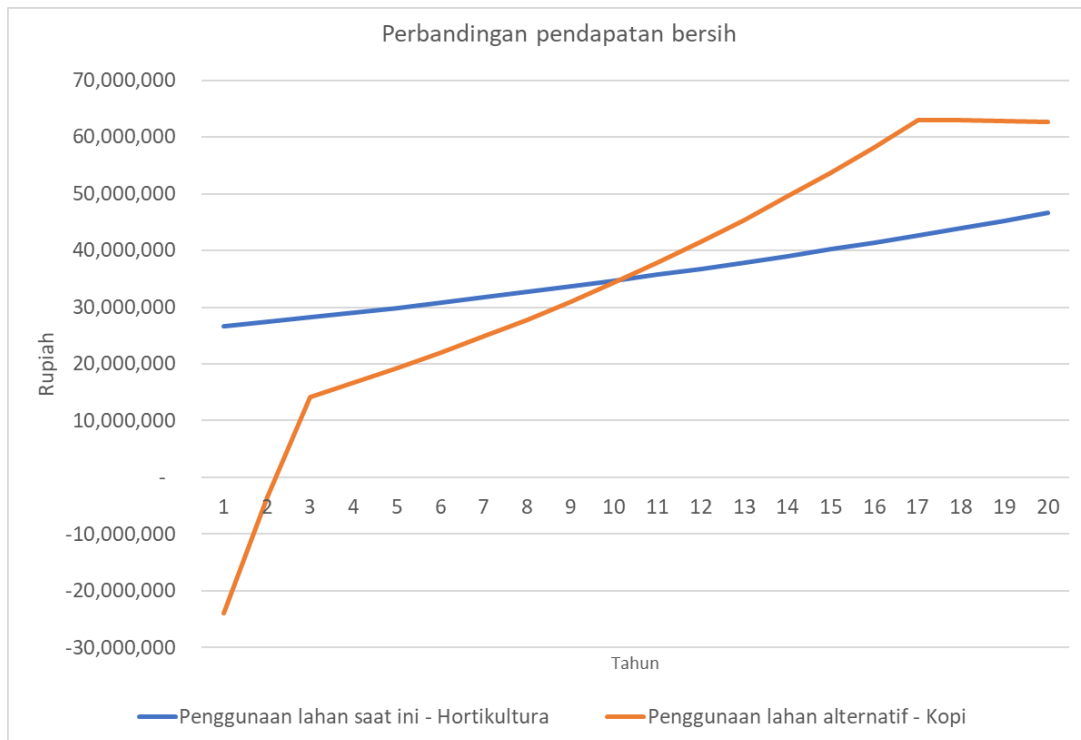
Comparison	Description
Average annual revenue	The average revenue per year over 20 years.
Average labour requirements	The average labour requirements per year over the 20 years.
Labour days to shift land use option	The number of labour days required to shift land use.
Average annual costs	The average costs per year over 20 years.
Average net revenue	The average net revenue (revenue minus costs) per year over 20 years.
Net present value over 20 years - with value of timber	The average net revenue (revenue minus costs) per year over 20 years – including the growth in timber value if timber trees are present.

The below snapshot shows an example of the information when changing from horticulture to coffee on steep sloping land.

As it can be seen in the first chart, horticulture provides greater returns initially, but coffee gradually overtakes horticulture as the preferred land use option. Total labour requirements are much lower (the second chart).

Despite this, the net present value of horticulture is still greater because of the relatively high annual returns and the high discount rate making cash flow in initial years relatively more important.

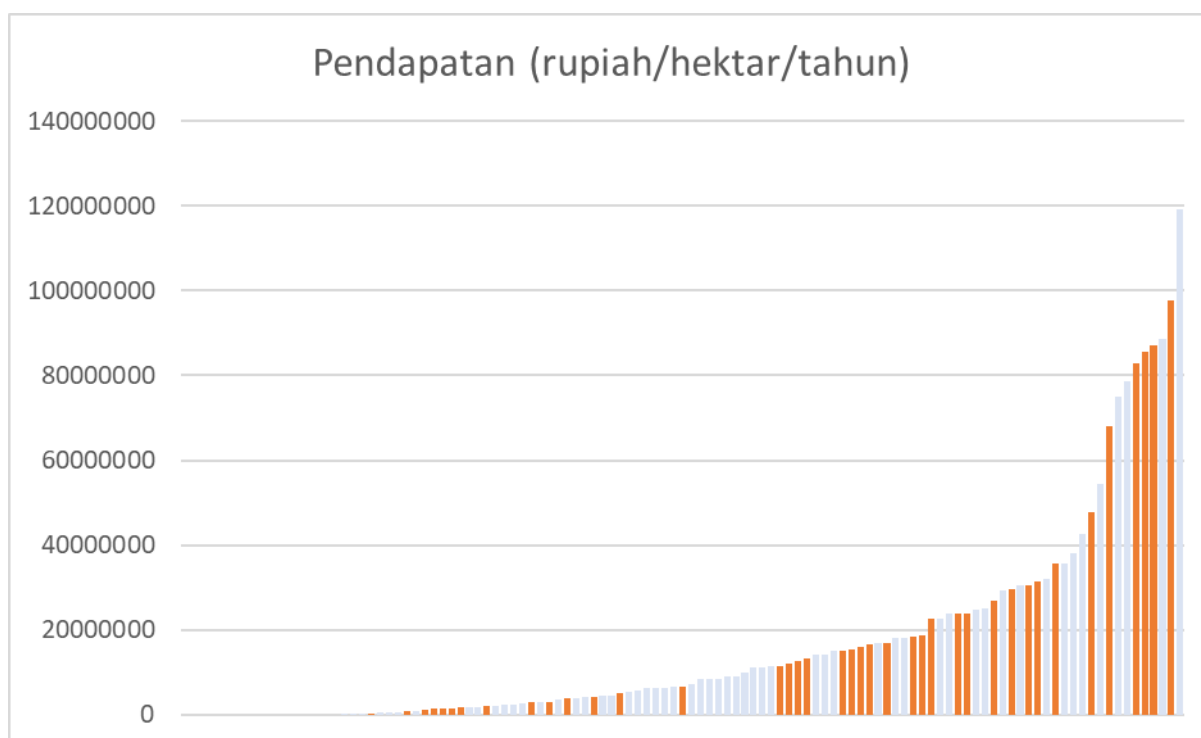
Pendapatan tahunan rata-rata		
Penggunaan lahan saat ini - Hortikultura	67,175,936	Rupiah
Penggunaan lahan alternatif - Kopi	47,314,498	Rupiah
Persyaratan Tenaga kerja rata-rata		
Penggunaan lahan saat ini - Hortikultura	340	Hari kerja
Penggunaan lahan alternatif - Kopi	136	Hari kerja
Hari kerja beralih antara Kopi dan di tahun pertama	174	Hari kerja
Biaya tahunan rata-rata		
Penggunaan lahan saat ini - Hortikultura	31,485,361	Rupiah
Penggunaan lahan alternatif - Kopi	12,309,889	Rupiah
Pendapatan bersih rata-rata		
Penggunaan lahan saat ini - Hortikultura	35,690,575	Rupiah
Penggunaan lahan alternatif - Kopi	35,004,609	Rupiah
Laba bersih selama 20 tahun		
Penggunaan lahan saat ini - Hortikultura	226,487,216	Rupiah
Penggunaan lahan alternatif - Kopi	129,296,933	Rupiah
Laba bersih selama 20 tahun - dengan nilai pohon kayu		
Penggunaan lahan saat ini - Hortikultura	226,487,216	Rupiah
Penggunaan lahan alternatif - Kopi	129,296,933	Rupiah



3 How to use - Perbandingan Pertanian – Kopi

The farm comparison sheet for coffee – or *Perbandingan Pertanian – Kopi* – allows for users to observe the real-life outcomes of coffee plots collected in the survey.

The survey collected detailed information of 113 plots that primarily grow coffee. The revenue, costs, net income, and labour requirements are ordered and plotted just like the chart below (for revenue):



This first provides an indication on the range input and output outcomes for coffee plots within the sample.

The user is then able to filter to see plots with particular characteristics, using drop down lists at the start of the sheet.

Daftar lereng

Praktek konservasi tanah dan lahan

Kopi dengan pohon kayu

Kopi dengan pohon buah-buahan

Kopi dengan komoditas lain musiman (Sayuran, Kacang-kacangan)

Gunakan pupuk

Gunakan bahan kimia

Pangkas pohon

Kepadatan pohon

[illegible]

The drop-down list options include:

- Plot slope (flat/moderate, and steep).
- Practices soil conservation.
- Coffee with timber trees.
- Coffee with fruit trees.
- Coffee with other annual commodities.
- Uses fertiliser.
- Uses chemicals.
- Whether the coffee trees were pruned in the last year.
- Tree density (lower than the median, higher than the median).

Note

To ignore the data filter option, the relevant cell can be left blank, or *skip data filter option* can be selected in the drop-down list.

The plots that match the criteria set by the user will be highlighted in orange.

For example, the user may be interested in coffee plots that practice soil conservation. They would set the following filter with the drop-down menu:

Daftar lereng

Praktek konservasi tanah dan lahan

Kopi dengan pohon kayu

Kopi dengan pohon buah-buahan

Kopi dengan komoditas lain musiman (Sayuran, Kacang-kacangan)

Gunakan pupuk

Gunakan bahan kimia

Pangkas pohon

Kepadatan pohon

Ya

The results show the following outcomes for profits:

Jumlah plot dengan karakteristik yang dipilih

45 dari 113 plot

Pendapatan Rata-rata per tahun

19,912,168 Rupiah / hektar / tahun

Rata-rata biaya tunai per tahun (termasuk biaya sewa tenaga kerja)

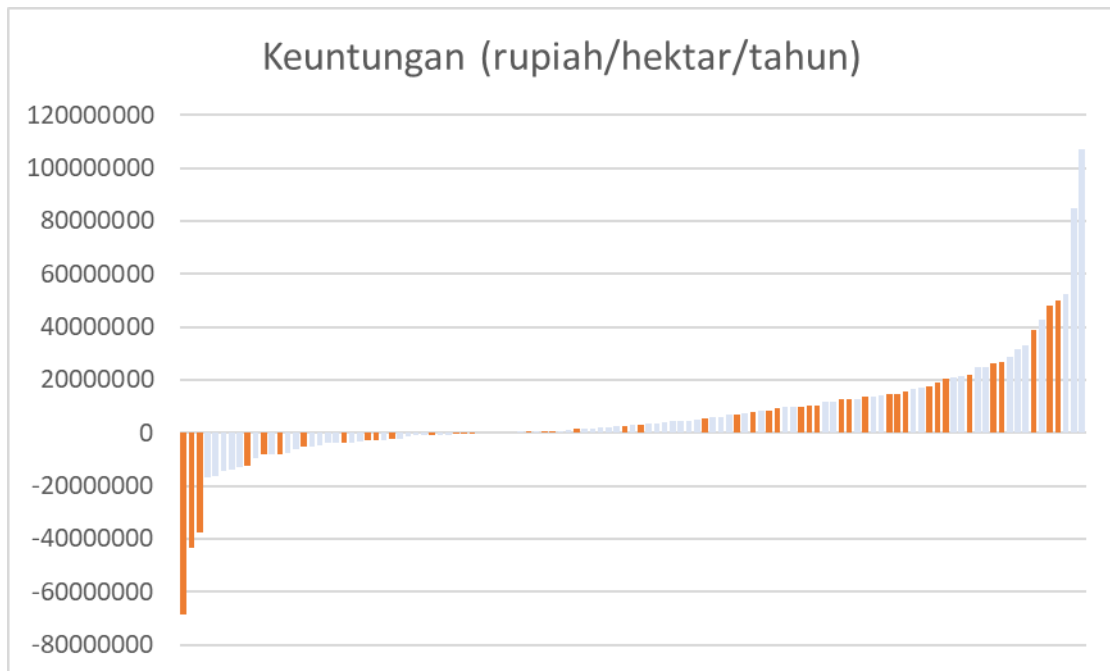
14,767,840 Rupiah / hektar / tahun

Keuntungan rata-rata (termasuk biaya tenaga kerja sewa)

5,144,328 Rupiah / hektar / tahun

Penggunaan tenaga kerja rata-rata per tahun (rumah tangga dan tenaga kerja sewaan)

207 Hari kerja / hektar / tahun



Insights from this filter indicate that soil conservation is possibly associated with higher revenues, but the costs are much higher. This results in some variation in overall profits, with some plots experiencing very low profits.

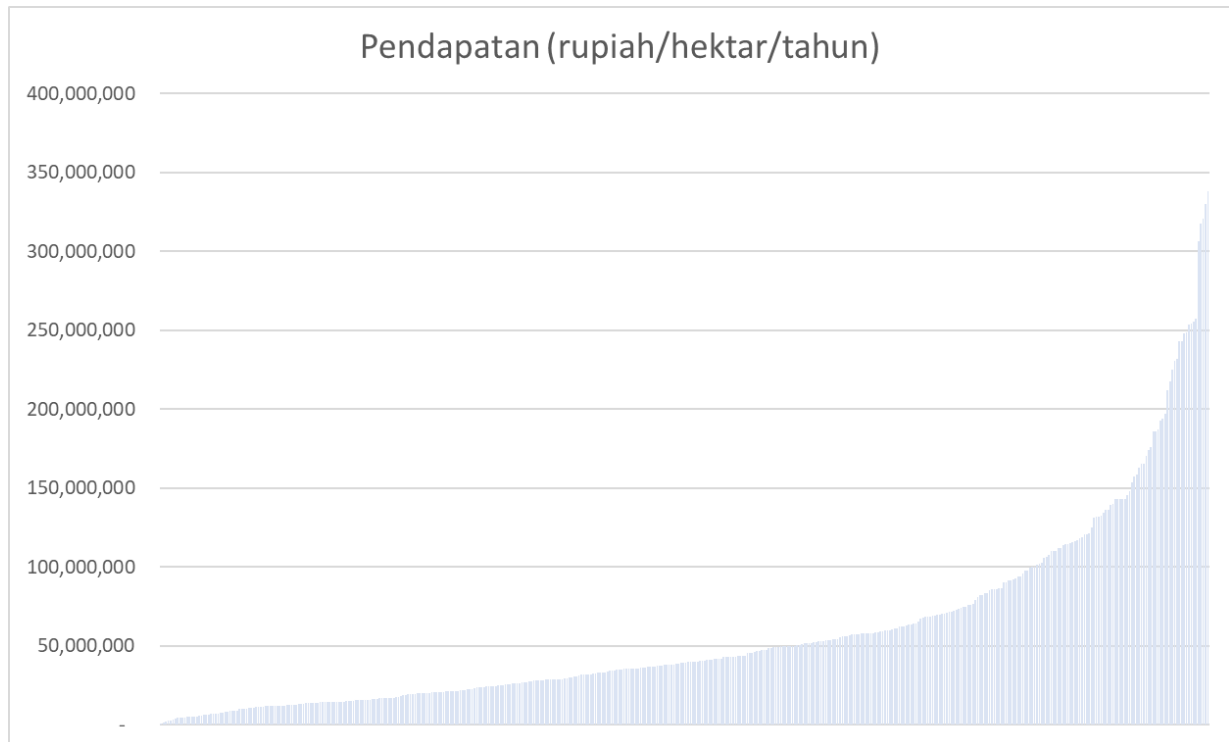
Note

There may be no plots that match the criteria chosen by the user, especially if a number of different filters are chosen.

4 How to use - Perbandingan Pertanian – lainnya

The farm comparison sheet for coffee – or *Perbandingan Pertanian – lainnya*– allows for users to observe the real-life outcomes of horticultural, corn or rice plots collected in the survey.

The survey collected detailed information of 442 plots that primarily grow coffee. The revenue, costs, net income, and labour requirements are ordered and plotted just like the chart below (for revenue):



This first provides an indication on the range input and output outcomes for plots within the sample.

The first option is to select a land use – horticulture, corn or rice. Plot slope and soil conservation can also be selected at this stage but can be ignored if the user wishes not to filter on these criteria.

Harap berikan musiman komoditas yang ingin Anda pertimbangkan?
Daftar lereng
Praktek konservasi tanah dan lahan

By doing this, new data filter options will become available depending on the land use. These filters include:

- For horticulture:
 - Number of harvests a year

- Horticulture with agroforestry
- Intercropping with other annual crops
- Main crop types
 - Chilli
 - Cabbage
 - Carrot
 - Potato
 - Tomato
 - Onion/shallots
 - Other vegetables
- For rice:
 - Number of harvests
 - Intercropping with vegetables
 - Rotate crops between seasons with
 - Corn
 - Vegetables
 - Tubers
- For corn
 - Number of harvests
 - Intercropping with other annual crops
 - Rotate crops between seasons with
 - Rice
 - Vegetables
 - Tubers

The plots that match the criteria set by the user will be highlighted in orange.

For example, the user may be interested in horticultural plots with chilli as the main crop type. They will filter to horticulture, then select chilli from the drop-down list.

Hortikultura	
Jumlah musim panen	
Hortiultur dengan agroforestri	
Tumpangsari dengan tanaman tahunan multipel lainnya	
Komoditas Utama	Cabe

There are 36 plots with chilli as the main crop. The following results suggest that chilli is relatively profitable for a few plots, although the costs can be high. Note that in this instance the median profit does not need to equal the median revenue minus

the median costs, as these median values are taken independent of each other from the data.

Jumlah plot dengan karakteristik yang dipilih

36 dari 442 plot

Pendapatan Rata-rata per tahun

33,244,048 Rupiah / hektar / tahun

Rata-rata biaya tunai per tahun (termasuk biaya sewa tenaga kerja)

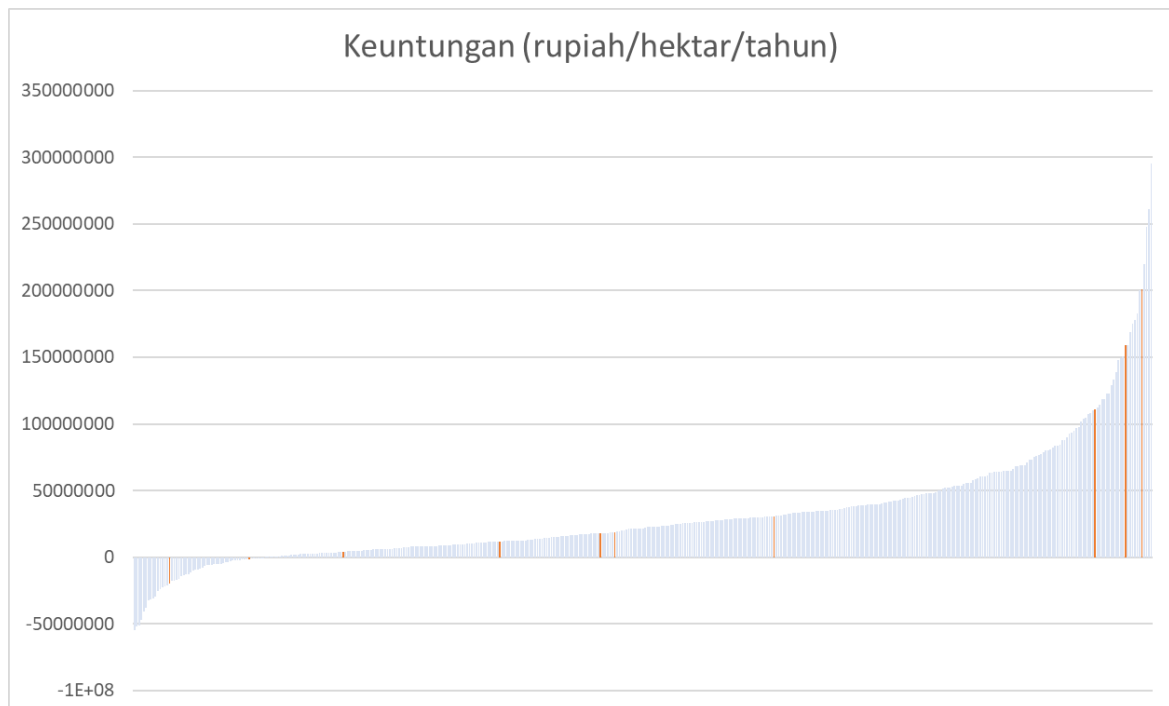
43,666,667 Rupiah / hektar / tahun

Keuntungan rata-rata (termasuk biaya tenaga kerja sewa)

18,196,427 Rupiah / hektar / tahun

Penggunaan tenaga kerja rata-rata per tahun (rumah tangga dan tenaga kerja sewaan)

442 Hari kerja / hektar / tahun



5 Other sheets

The excel based model has a number of additional sheets that are hidden from the user. These can be unhidden and edited by users if required by right-clicking on any visible sheet, clicking on “unhide”, and selecting a relevant sheet.

5.1.1 Input sheets

Firstly, there are four input sheets – coffee, rice, horticulture and mixed agroforestry. These sheets take the inputs from the dashboard, and contain all other inputs and assumptions required for the modelling.

The input sheets are designed to be hidden and not changed between uses. In some cases it may be required to change some inputs that cannot be changed on the dashboard, due to an error or updated information.

To change any values on the input sheets, first save a new version and then unhide the input sheet of interest. Inputs that are in grey cells like those below can be changed. Those in white cells should not be changed as these are formulas and are required for the modelling.

Biaya masukan	Masukan aktif	Datar	Sedang	Curam
Pupuk	250000	110000	110000	250000
Pestisida dan herbisida	215	215	215	215
Biaya bibit (kopi)	2000	2000	2000	2000

Note

Detailed information on data sources and assumptions can be found in the input sheets of the excel spreadsheet, and more data and assumptions are detailed in Section 4.2.

5.1.2 Other hidden sheets

There are a range of other hidden sheets that should not be changed. These include:

- **Models** – There are four model sheets – coffee, rice, horticulture and agroforestry. These model sheets take the inputs from the dashboard and the input sheets and estimate the revenues and costs for presenting on the dashboard. These sheets should not be modified.
- **Data – annual crops**. These sheets (revenue, costs, labour and profits) contain the data used for the annual crop farm analysis tabs. These sheets should not be modified.
- **Kopi analysis**. These sheets (revenue, costs, labour and profits) contain the data used for the coffee farm analysis tabs. These sheets should not be modified.

- **Data for export** – These sheets are used to export the data used for other modelling purposes for the ACIAR project ADP/2015/043. These sheets should not be modified.
- **Daftar** – The text lists used in the drop-down lists. These should not be modified unless there are typos or errors in the drop-down lists.
- **Timber production profiles** – A sheet to profile timber growth. This sheet should not be modified.

6 Data and assumptions for the dashboard (Dasbor)

In this section the assumptions and data used to inform the results on the Dashboard (Dasbor) are outlined for each crop type – coffee, rice, horticulture, fruit trees, avocado, cinnamon, and timber.

6.1 Coffee

Coffee inputs are used for the coffee and are also used in the mixed agroforestry land uses where the user indicates coffee as a crop within the mixed agroforestry system.

In total, there are 182 plots in the ACIAR Survey ADP043 BANDUNG 2019 that have coffee as their main crop. These plots may grow coffee in a monoculture system or have a mixed coffee system with other crops and timber trees.

6.1.1 Coffee outputs

The modelling of coffee yields requires an understanding of coffee yields for plots over a 20-year period, including how the yield may increase or decline as the age of the tree changes. This is difficult to derive from the Bandung survey given the survey is cross-sectional and did not collect information of average ages of coffee trees. Other limitations of the coffee yield information in the data are presented in the appendix.

A yield gap methodology is used to model yields of coffee trees. As the modelling is not considering the marginal contributions of other inputs and environmental factors, the only factor that determines the distance between maximum attainable yield and current yield is the current age of the trees. This is represented in the equation below, where β_t represents the yield adjustment for a given tree age t .

$$Y_{coffee} = \beta_t Y_{potential}$$

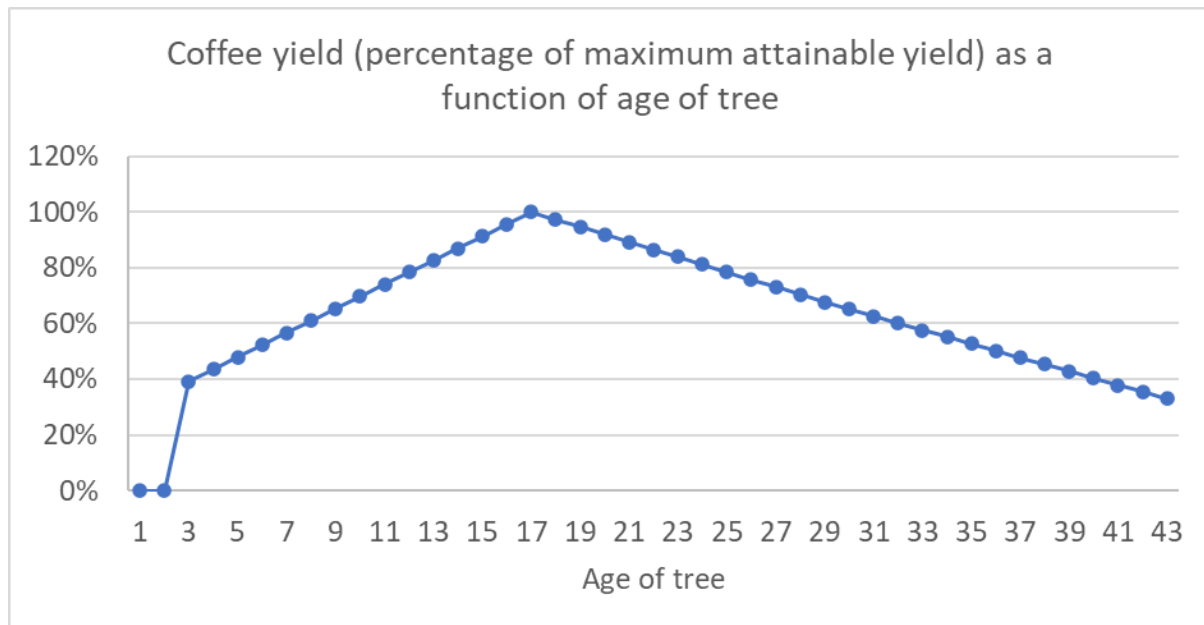
The yield adjustments as a function of tree age are derived from Wang et al (2015) and Lisnawati et al (2017). Lisnawati et al (2017) provides detailed tree and yield growth for a similar context to Bandung (southern Sulawesi smallholder farmers) for the first 17 years, while Wang et al (2015) provides the maximum attainable yield decline from 20 years up to 45 years (where it will be assumed the trees need to be replanted). A more detailed literature review of coffee yield profiles is provided in the appendix.

For the first 17 years, the percentage of maximum attainable yield is derived based on the equations presented in Lisnawati et al (2017). These equations are linear

approximations from the data that relate coffee tree diameter to age, and then diameter to coffee yields.

For the remaining 20 years, a linear approximation from Wang et al (2015) is be used, assuming a decline in yield to 28% when the tree is to be replanted.

The resulting yield adjustments as function of tree age is shown in the figure below, with the maximum attainable yield being reached by age 17 and declining thereafter.



The maximum attainable yield is represented as the 85th percentile of observed coffee yields, which is a typical quantile chosen in yield gap analysis. The maximum attainable yield per tree is higher in steep slopes as they are typically at higher altitudes more suitable for arabica production. The maximum attainable yields are shown below and are within what is expected for Arabica trees in the region (Lisnawati *et al.* 2017).

- Flat/moderate slopes – 2.04 kg per tree
- Steep slopes – 2.796 kg per tree

The Bandung Household Survey indicates a low number of productive trees per hectare – around 50% of total trees. This could be for a range of reasons such as old trees, young trees not yet mature, stumping, poor condition, disease or poor growing practices.

Given it is difficult to determine why households have few productive trees, two scenarios can be considered within the model. The default scenario assumes a normal, productive coffee production system where all trees are productive (subject to their age). The second scenario allows the user to input the percentage of productive per hectare within the Bandung Household Survey. The number of trees

per hectare is summarised in the table below. Harvest and post-harvest labour are scaled relative to the number of trees.

Number of coffee trees per Ha

	Flat	Moderate	Steep
Default number of coffee trees per hectare (assumed scenario)	2381	2381	2142
Median percentage of productive trees in the survey.	~65%	~65%	~55%

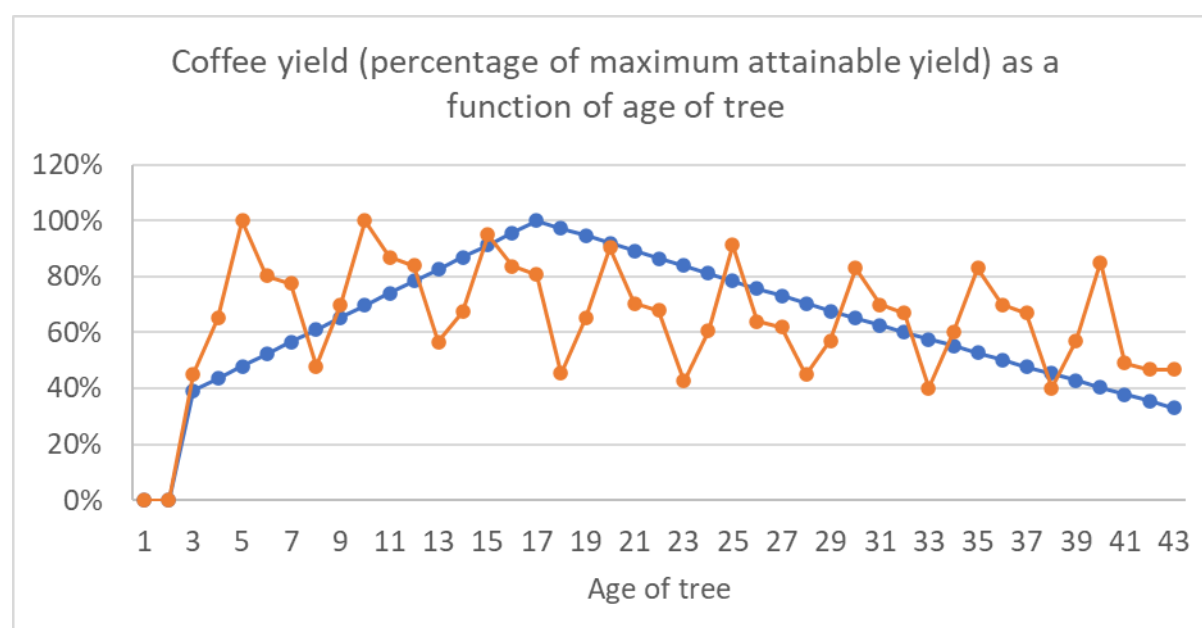
As a calibration exercise the median and mean yields from the Bandung Household survey are compared to the estimated yields. If it is assumed that there is an equal likelihood of a coffee tree being of any age between 0-44, and assuming the number of productive trees at around 55%-65%, the mean and median are only slightly higher than the mean and median values reported in the dataset. This is intuitive given that the data likely contains a higher number of older, less productive trees.

Outputs for coffee may be affected by a number of other agronomic, environmental and management practices (Mokondoko *et al.* 2022). Firstly, Anhar *et al* (2021) finds that shade trees (after a density of 100 trees a hectare) increase coffee yield per tree by 10.5%, but only if the shade trees are sufficiently established (e.g. >10 years). It is assumed that this uplift is already considered within the yield potentials of the Bandung survey given many plots have mixed agroforestry systems within their coffee plots.

Finally, the profile of yield does not include and variation in yield due to pruning, stumping or biennial production. As described in Somarriba and Quesada (2022), pruning and stumping can be done staggered across the plot to minimise fluctuations in yield, and biennial yields of individual trees may smooth out across a whole plot.

The graph below overlays the profile outlined in this section with that from Schreinemachers and Berger (2011). This profile was used in a previous version of the excel cash flow model. Production over the 40 years is approximately similar, meaning that the smoother yield profile is preferred as it can be considered an approximation of yield where there are unpredictable biennial yields.

Coffee yield (percentage of maximum attainable yield) as a function of age of tree- comparison with pruning cycle assumptions



6.1.2 Coffee inputs

Two stages of a coffee plot lifecycle are considered when analysing inputs and labour for coffee plots - coffee gardens that are newly established and not yet productive, and productive gardens. For newly established coffee gardens, the coffee plots were filtered to only consider plots with no productive coffee trees and where coffee seedlings were purchased in the last year. This provides a small sample of ten plots which is considered insufficient. For the model, it is assumed that fertiliser costs, chemical costs, and labour for weeding, fertiliser and chemicals, is the same as fully productive plots.

From these newly established plots however, the price of coffee trees seedlings could be found to be 2000 rupiah per seedling. For new coffee plots, this seedling cost is allocated in the first year, with a 90% survival rate.

For fully productive coffee plots, the sample of 182 coffee plots is filtered to only consider plots that are primarily used for coffee and are relatively productive. The filtering criteria include:

- Indicate coffee as the main crop.
- Have at least 500 productive coffee trees per Ha.
- Indicate that at least 50% of these trees are productive.

This leaves 5 flat plots, 51 moderate slope plots, and 20 steep slope plots. For the model, it is assumed inputs and outputs are the same for flat and moderate slope plots given the small sample of flat coffee plots. The table below outlines the inputs and outputs assumed for flat/moderate sloped coffee plots and steep coffee plots.

Inputs are derived from the median values in the Bandung household survey, and the outputs summarise the assumptions outlined in the previous section.

Input and output parameters for coffee plots

Variable	Flat slopes (56 observations)	Moderate/steep slopes (20 observations)
Labour – land preparation (labour days/ha)	0	0
Labour – planting (labour days/per tree)	0.028	0.028
Labour – weeding (labour days/ha)	35	23
Labour – fertiliser (labour days/ha)	10.5	13
Labour – chemicals (labour days/ha)	7.1	7.1
Labour – harvest (labour days/per tree)	0.0476 * yield adjustment	0.0476 * yield adjustment
Labour – post harvest (labour days/per tree)	0.0025 * yield adjustment	0.0025 * yield adjustment
Fertiliser costs (Rupiah/per ha)	110,000	250,000
Chemical costs (Rupiah/per tree)	215	215
Seedling cost (Rupiah/per seedling)	2,000	2,000
Yield potential per tree (kg/tree)	2.04	2.796
Average age of established coffee trees (years)	15	15

Harvest and post-harvest labour are scaled relative to the production of the tree, which is determined by the age of the tree as outlined in the previous section. Land preparation labour is equal to zero once the plot is established as there is no need to maintain or re-establish the soil and beds after harvest. Planting labour is only relevant in the first year where there are new coffee seedlings.

The default, median coffee price from the Bandung household survey is 8,000 rupiah per kg of coffee cherries. This can be edited by the user on the Dashboard. The average age of the tree can also be overridden in the Dashboard.

6.2 Horticulture plots

Horticulture plots include all plots that have indicated vegetables or similar annual crops as their main crop type. In total 332 plots were identified as having horticulture as their main crop. The survey collected information of up to three different crops on the plot at any one time, for up to three seasons in the year. As shown the table below, Chilli, potato and cabbage crops are the most frequently grown crops, although there is significant variety in the types of crops grown across seasons. Just over 40% of plots have multiple crops grown simultaneously, again with chilli and cabbage being the most frequently grown secondary and third crops. Just over half of the plots have one harvest season a year, with 29% having two harvest seasons and the remainder three harvest seasons.

Crop types observed for horticulture in the Bandung Household Survey

Crop type	Number of plots indicating main crop
Chilli	55
Cabbage	44
Potato	42
Tomato	38
Carrot	32
Shallot	19
Green onion	17
Broccoli	14
Green beans	11
Other Vegetables	60

To account for the multiple seasons and multiple crops grown simultaneously across seasons, inputs and outputs were aggregated across the year and crops to provide annual inputs and outputs for all crops. This makes inputs and outputs comparable with coffee and mixed systems where outputs and inputs are for one harvest season a year.

For outputs, total production of crops is used to account for produce consumed by the household. The table below details annual inputs and outputs for the horticulture plots. For all inputs and outputs, the median value from all horticulture plots is used, for the given plot slope.

Input and output parameters for horticulture plots

Variable	Flat slopes (110 observations)	Moderate slopes (164 observations)	Steep slopes (58 observation)
Labour – land preparation (labour days/ha)	50	59	89
Labour – planting (labour days/ha)	31	31	42
Labour – weeding (labour days/ha)	47	42	71
Labour – fertiliser (labour days/ha)	35	30	58
Labour – chemicals (labour days/ha)	31	41	48
Labour – harvest (labour days/ha)	33	33	24
Labour – post harvest (labour days/ha)	1	1	3
Fertiliser costs (Rupiah/per ha)	1,700,000	1,500,000	950,000
Chemical costs (Rupiah/per ha)	1,800,000	3,500,000	2,000,000
Seed costs (Rupiah/per ha)	85,000	85,000	85,000
Revenue per Ha (Rupiah/per ha)	67,000,000	54,000,000	31,000,000

Labour for fertiliser and weeding initially declines from flat to moderate slopes, before increasing for steep slopes. As for chemicals such as pesticides and herbicides, total costs initially increase from flat to moderate slopes before decreasing for steep slopes. For steep slopes this is intuitive given the high labour input requirements for steep sloping land being offset by less intensive production. Lower weeding and fertiliser labour for moderate slopes can be explained by the higher herbicide use on moderate slopes.

The model requires both production in kilograms and a price, rather than total revenue. For horticulture plots with multiple outputs, a weighted average price for all

outputs is derived from the horticulture plots of 3000 rupiah per kg. Total production is derived by dividing the median revenue by the weighted average price.

6.3 Rice plots

In total, 204 plots have rice as the main crop in the Bandung survey and have detailed data. 67 of these plots are on flat sloped land, 125 are on moderate sloped land and the remaining 12 are on steep sloping land.

Rice plots are expected to have some diversity in the survey data, with flatter plots having more paddy plots, with steeper plots more likely to have dryland rice intercropped or rotated with other annual crops such as maize and vegetables.

The majority of rice plots have multiple seasons, with a dry season harvest and a wet season harvest the most common (58% of total plots). Other plots either have one harvest a year, or up to three harvests a year. To account for the multiple seasons, annual inputs and outputs for rice are calculated, rather than seasonal outputs and inputs, to be comparable coffee and mixed systems where outputs and inputs are for one harvest season a year.

For outputs, the yield from rice is calibrated based on the total on farm revenue. This was achieved by estimating the total value of production across all commodities on each plot, across seasons, commodity types and accounting for household consumption. Given a median price of rice of 6,000 rph, a median yield of rice per hectare was estimated by dividing the median total revenue per hectare by the median price for rice. While there was flexibility to include other commodities such as vegetables in the data, the majority of plots only have paddy or dryland rice, so having only rice as an output in the model is representative of the majority of rice plots.

The table below details annual inputs and outputs for the rice plots. For all inputs and outputs, the median value from all rice plots is used, for the given plot slope.

Input and output parameters for rice plots

Variable	Flat slopes (67 observations)	Moderate slopes (125 observations)	Steep slopes (12 observations)
Labour – land preparation (labour days/ha)	82	98	124
Labour – planting (labour days/ha)	50	60	60
Labour – weeding (labour days/ha)	130	100	135

Labour – fertiliser (labour days/ha)	15	20	20
Labour – chemicals (labour days/ha)	38	50	38
Labour – harvest (labour days/ha)	45	45	45
Labour – post harvest (labour days/ha)	13	15	15
Fertiliser costs (Rupiah/per ha)	90,000	250,000	250,000
Chemical costs (Rupiah/per ha)	350,000	430,000	430,000
Seed inputs (Kg/per ha)	50	50	50
Yield per Ha (Kg/per ha)	7,600	6,380	3398

For pesticides, the mean value of 350,000-430,000 rupiah per hectare is used given the median value is zero. Many of the plots do not use pesticides or fertilisers, although the ones that do allocate significant labour and resources to spraying. There may be trade-offs associated with labour inputs such as weeding and chemical applications depending on capital availability. Median labour requirements for both spraying and weeding are used for all plots, noting that foregoing one activity will likely increase the requirements for the other significantly.

For seed costs, 50kg of rice seed per hectare is assumed based on Connor *et al* (2021). This is multiplied by the price of rice per kg, derived from the survey as 6,000 rupiah per Ha.

Input costs are likely to be much higher for steep slopes due to the requirements for terracing or other conservation practices to maintain the beds. This is captured in the median values, and in the land establishment cost when transitioning from other land uses.

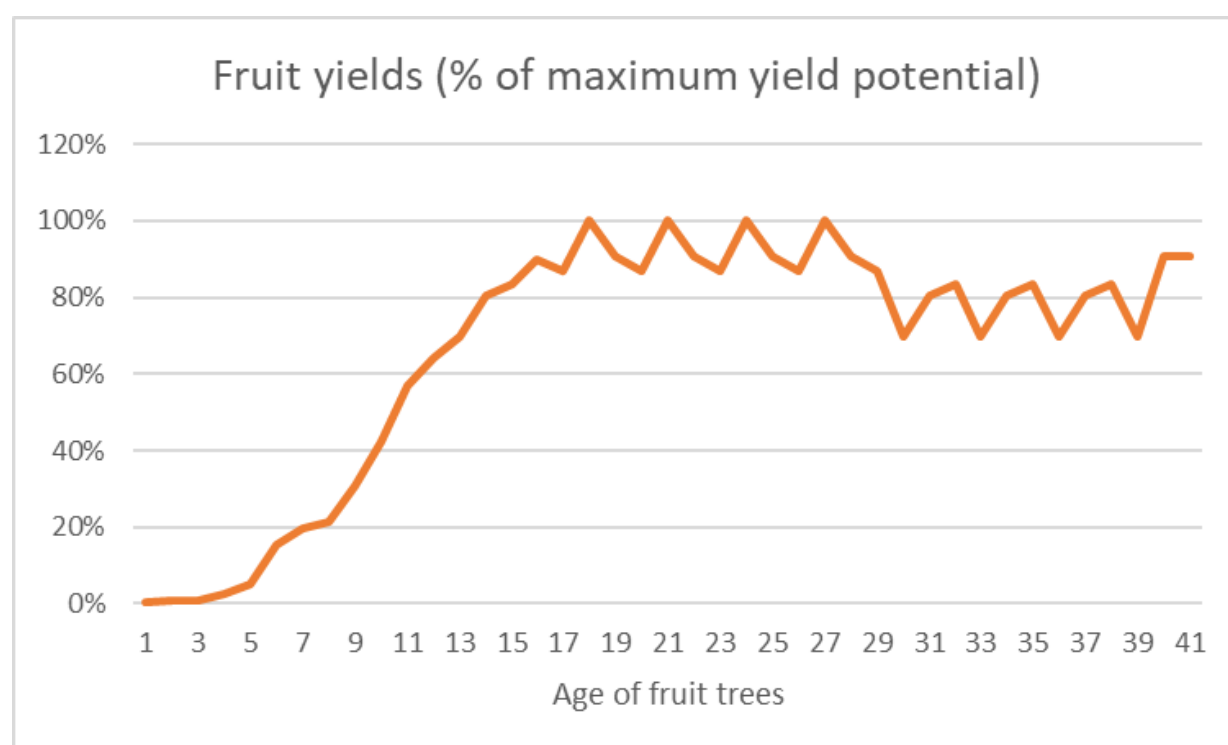
6.4 Fruit trees

6.4.1 Fruit tree outputs

Fruit production follows a similar methodology to coffee, the yield from fruit trees is a function of the current age of the trees and the maximum attainable yield. The maximum attainable yield is the 80th percentile in fruit revenues per tree from the Bandung household survey, equal to 28,750 rupiah per tree per Ha.

A weighted average price from the survey of 2,500 rupiah per kg is used to estimate a weighted average production in kg.

The yield adjustment based on the age of the tree is derived from production profiles provided to the INDOGREEN project by ICRAF, which considers inputs and outputs for fruit-based coffee systems in West Lampung Indonesia. There are a variety of fruits available for mixed agroforestry systems, with some reaching production maturity earlier than others. A weighted average mix of nine different fruit trees was used to determine the percentage of production based on the age of the agroforestry system and is plotted in the figure below.



The yields fluctuate after year 17 due to natural biennial variation in fruit tree production such as jackfruit, durian and rambutan.

6.4.2 Fruit tree inputs

Given inputs for mixed agroforestry fruit tree systems are reported for the entire plot, and not for each crop, it is difficult to derive individual input requirements for fruit and timber trees. Assumptions have been made based input aggregate input requirements for the entire plots and are summarised in the table below. It is assumed that many of these inputs are the same as with coffee, albeit with less weeding for fruit trees, and no pests and herbicides. Pest and herbicide labour is still included as it is assumed applications take longer per tree for coffee trees within agroforestry systems.

As with coffee, there are initial upfront costs associated with planting fruit and timber trees. The labour requirements and cost of seedlings are higher than coffee trees (20,000 rupiah/seedling) as the fruit tree seedlings tend to be more mature. This

was validated with interviews by the research team in the field with farmers in November 2022.

Input and output parameters for mixed agroforestry plots – fruit trees

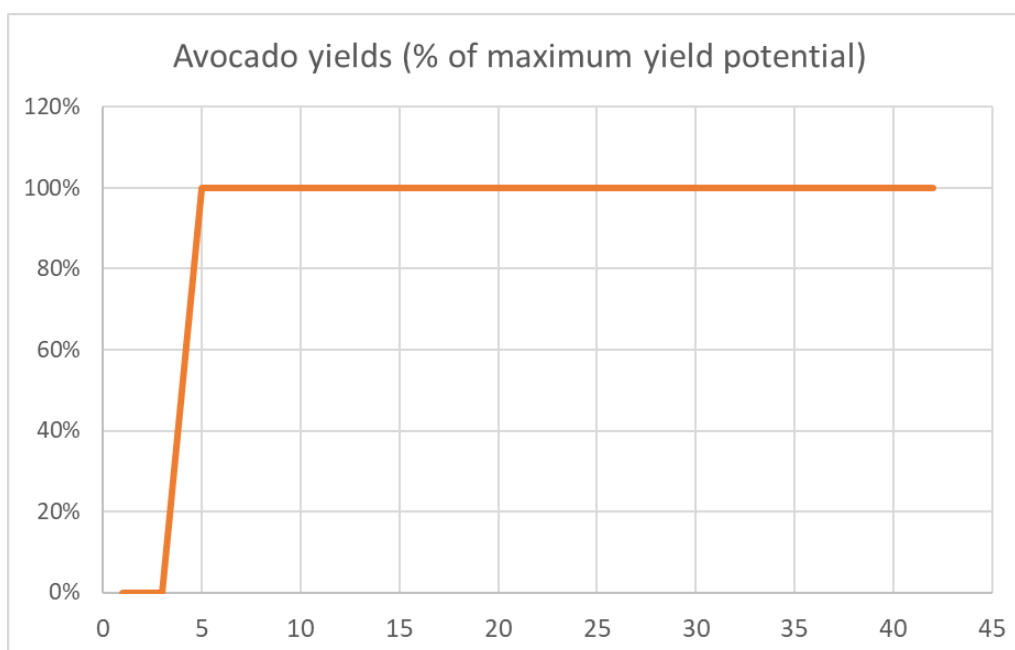
Variable	Flat/moderate slopes	Steep slopes
Labour – land preparation (labour days/ha)	0	0
Labour – planting (labour days/tree)	0.028	0.028
Labour – weeding (labour days/ha)	15	12
Labour – fertiliser (labour days/ha)	10.5	13
Labour – chemicals (labour days/ha)	7.1	7.1
Labour – harvest (labour days/tree)	0.0476 * yield adjustment	0.0476 * yield adjustment
Labour – post harvest (labour days/ha)	0.0025 * yield adjustment	0.0025 * yield adjustment
Fertiliser costs (Rupiah/per ha)	251,275	448,674
Chemical costs (Rupiah/per tree)	0	0
Seedling inputs (Rupiah/per seedling)	20,000	20,000
Yield potential per tree (Rupiah/tree)	28,750	28,750
Default age of established fruit trees (years)	15	15

6.5 Avocado

6.5.1 Avocado outputs

Avocado outputs have been estimated based on secondary sources, rather than the household survey. This is due to limited sample size for avocado trees independent of other crops such as coffee and other fruit trees.

A yield potential approach based on the age of the tree is used for avocado trees, as with coffee and fruits. The yield potential for all slopes is set at 6kg per tree, following interviews with producers in November 2022. This is the mean yield indicated, as no applicable data was found to provide the yield as a function of the age of the avocado tree (which would otherwise be used to adjust the actual yields down). The trees take 5 years to become fully productive, with 50% of the trees productive in the 4th year.



6.5.2 Avocado inputs

Avocado tree inputs either assume the same input requirements for other fruit trees or have been validated from interviews with avocado growers. Avocado trees require a significant amount of pruning labour per tree to maintain productivity, which is assumed to be 200 labour days per hectare (approx. ten trees per day can be pruned, spread over 4 days a year).

Seedling costs are higher than fruit trees at 50,000 rupiah per seedling. This was validated with interviews by the research team in the field with farmers in November 2022.

Note that measures on a per hectare basis assume a maximum of 2330 trees for flat slope plots and 2142 trees for steep slope plots.

Input and output parameters for mixed agroforestry plots – avocado trees

Variable	Flat/moderate slopes	Steep slopes
Labour – land preparation (labour days/ha)	0	0
Labour – planting (labour days/tree)	0.028	0.028
Labour – weeding and pruning (labour days/ha)	200	200
Labour – fertiliser (labour days/ha)	17.5	21
Labour – chemicals (labour days/ha)	7.1	7.1

Labour – harvest (labour days/tree)	0.095 * yield adjustment	0.095 * yield adjustment
Labour – post harvest (labour days/tree)	0.0025 * yield adjustment	0.0025 * yield adjustment
Fertiliser costs (Rupiah/per ha)	1.2 million	1.2 million
Chemical costs (Rupiah/per tree)	333	333
Seedling inputs (Rupiah/per seedling)	50,000	50,000
Yield potential per tree (Kg/tree)	6kg	6kg
Default age of established fruit trees (years)	15	15

6.6 Cinnamon

6.6.1 Cinnamon outputs

Cinnamon is typically harvested in 3-6 year cycles, with the tree growing new branches from the stump after each harvest until completely replaced (Rositiana *et al* 2020). For the model, it is assumed a harvest occurs every 5 years, with the harvest amount not changing over the life of the tree. This avoids the need to have the age of the cinnamon trees and yield functions based on the age of the tree.

The yield per tree is assumed to be 4.982 kg per tree based on Rositiana *et al* (2020). The default price is assumed to be 2000 rupiah per kg of wet, unprocessed cinnamon but can be changed by the user on the dashboard.

6.6.2 Cinnamon inputs

The labour and capital inputs are summarised in the table below. Much of the inputs are similar to other tree crops, although harvest labour and post harvest labour are slightly higher compared to other tree crops such as coffee and fruits.

Input parameters for mixed agroforestry plots – cinnamon trees

Variable	Flat/moderate slopes	Steep slopes
Labour – land preparation (labour days/ha)	0	0
Labour – planting (labour days/tree)	0.028	0.028
Labour – weeding (labour days/ha)	15	12
Labour – fertiliser (labour days/ha)	10.5	13

Labour – chemicals (labour days/tree)	7.1	7.1
Labour – harvest (labour days/tree)	0.04	0.04
Labour – post harvest (labour days/ha)	2	3
Fertiliser costs (Rupiah/per ha)	251,275	448,674
Chemical costs (Rupiah/per tree)	0	0
Seedling inputs (Rupiah/per seedling)	20,000	20,000
Average age of established cinnamon trees (years)	10	10

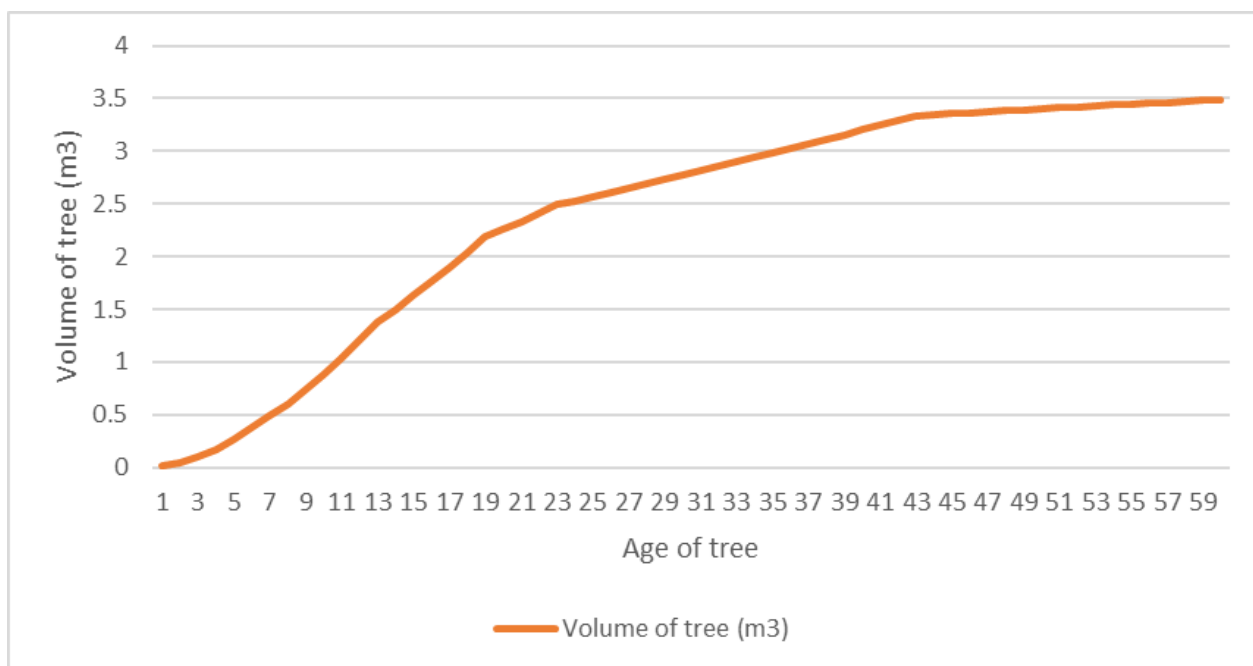
6.7 Timber

6.7.1 Timber outputs

Timber production is lumpy, and the timing of the harvest depends on the preferences of the household. For this reason, the data for the model does not consider cash flows from timber, but the volume of timber left standing in the plot. This volume can be converted to a value to make it comparable with other land uses.

The growth of timber volume is based on Krisnawati et al (2011). The Bandung household survey indicates a variety of timber trees across the sample, with the most common being Silk Trees, Eucalyptus, Umbrella trees and Burr-flower. Krisnawati et al (2011) detail growth rates of Burr-flower for the West Java region, and it is assumed these growth rates, and value from this timber, is considered representative of the timber options available for smallholders.

The chart below details the volume of timber per tree based on the age of the tree – derived first by estimating the diameter of the tree as a function of age, then determining the average height and volume of tree based on the diameter.



An average price per m3 of timber is from Irawanti et al (2017), equal to 875,000 per m3. The price per m3 of timber tends to increase with the height of the tree, and for this reason a median price based on the minimum and maximum per m3 from Irawanti et al (2017) is chosen.

The cost of harvesting timber is assumed to be equal to 7,358 rupiah per tree, assuming typical labour requirements to harvest timber trees. This cost is subtracted from the total value to provide a net value per m3 of timber. The time to remove stumps and replant trees is included in land preparation labour.

6.7.2 Timber inputs

Timber inputs assume similar costs to coffee, noting there will be some diversity in how timber plantations are managed by smallholders. Fertiliser and chemical costs are included, although in many systems these might be unlikely. The addition of these costs is due to most timber plots being intercropped with other crops such as coffee and fruits, where applications may be applied more generally on the plot, or at the very least the presence of timber plots reduces the efficiency of application per tree for the other crops.

Input parameters for mixed agroforestry plots – timber trees

Variable	Flat/moderate slopes	Steep slopes
Labour – land preparation (labour days/ha)	0	0
Labour – planting (labour days/tree)	0.028	0.028
Labour – weeding (labour days/ha)	0	0

Labour – fertiliser (labour days/ha)	10.5	13
Labour – chemicals (labour days/ha)	7.1	7.1
Labour – harvest (labour days/ha)	0	0
Labour – post harvest (labour days/ha)	0	0
Fertiliser costs (Rupiah/per ha)	251,275	448,674
Chemical costs (Rupiah/per tree)	333	333
Seedling inputs (Rupiah/per seedling)	20,000	20,000
Average age of established timber trees (years)	10	10

6.8 Other modelling assumptions

6.8.1 Cost of transitioning between land uses

Changing between land uses is assumed to have additional land preparation costs associated with clearing land, removing excess materials, and establishing beds and related land improvements.

It is assumed that labour is the only transition cost, given most the work is labour intensive in clearing land and moving soil. Barbier (1990) outlines labour requirements for soil conservation activities in uplands of Indonesia. While related to soil conservation and a dated study, the labour requirements are considered relevant due to similar technologies used for land use transition (labour intensive with manual tools) and the similar activities between soil conservation activities such as terracing and transitioning between land uses. Adiningsih and Karama (1992) find similar labour requirements for soil conservation activities.

The table below outlines the labour costs associated with transitioning between land uses for flat, moderate and steep slopes. Transitioning from horticulture and rice to other land uses assumes labour is required to move soil and re-do beds of between 150-200 labour days. Transitioning from coffee and mixed agroforestry to horticulture and rice systems has much higher labour requirements to include timber harvest costs, stumping, removing of material, new soil, and the construction of beds, and channels.

Moving from mixed agroforestry systems to coffee has slightly lower labour requirements as it is assumed the coffee trees will remain and reduce the need to introduce soil and establish new channels or beds. The same is true when transitioning from coffee to mixed agroforestry. This also has an impact on revenues

and inputs, where the plots have a mix of old and new coffee trees depending on the number of trees the current land use has.

In all cases it is assumed that labour requirements are higher on steep slopes. Costs for planting seedlings or seed costs are considered separately.

Land use transition labour requirements

Current land use	New land use			
	Coffee	Horticulture	Rice	Mixed agroforestry
<i>Flat plots</i>				
Coffee	0	292	292	76
Horticulture	150	0	150	150
Rice	150	150	0	150
Mixed agroforestry	226	292	292	0
<i>Moderate slope plots</i>				
Coffee	0	476	476	76
Horticulture	150	0	150	150
Rice	150	150	0	150
Mixed agroforestry	226	476	476	0
<i>Steep slope plots</i>				
Coffee	0	595	595	76
Horticulture	200	0	200	200
Rice	200	200	0	200
Mixed agroforestry	276	595	595	0

For plots that transition into horticulture or rice, it is assumed that there is a missed season of revenues in the first year (reducing horticulture revenues by half and rice by 33%). Harvest and planting inputs are adjusted down to reflect this missed season, although by a lesser amount to account for a learning period in the first year.

6.8.2 Timber revenue from transitioning from timber

If the land use transitions from a mixed agroforestry system to another system without timber trees, it is assumed the existing timber trees are harvested and sold. This is represented as revenue in the first year based on the number of timber trees harvested. For the model it is assumed the average age of these trees is 10 years. The costs associated with this harvest are captured in the land use transition costs.

6.8.3 Transitioning between different agroforestry systems

When transitioning between agroforestry systems, a few

Firstly, the user can specify whether the scenario changes the number of trees changes. If the number of coffee, timber, fruit, avocado or cinnamon trees changes between land uses, then the model will assume:

- If the number of trees increases, then it will be assumed that the old trees remain, and the remainder will be new seedlings.
- If the number of trees decreases, the remaining trees will be assumed to be old.

This has yield implications, where the older trees continue to produce yields and the new trees mature. It also has implications on the plot establishment costs, which for trees is assumed to be proportional to the number of trees that are removed and planted.

The user can also enter a scenario where the existing trees are stumped/heavily pruned for coffee. This just assumes the trees are replanted.

6.8.4 Picking ripe, red coffee cherries

Picking ripe, red coffee cherries is a harvest strategy to increase the quality of Arabica coffee and may attract a price premium in certain coffee value chains. The user can enter this harvest strategy on the dashboard, and in the model, it is assumed that this harvest strategy increases harvest labour by 30%.

The farmer may be compensated by receiving a higher price, which must be entered on the dashboard, otherwise the model assumes the same price.

6.8.5 Default prices

The model treats all outputs in physical quantities, in which is multiplied by a price for costs and revenue. The table below summarised all relevant prices for the model.

Summary of prices for outputs and inputs

	Price (rupiah) and unit	Comments and source
Coffee	8,000 / kg coffee cherries	Average price observed in Bandung Household Survey

Horticulture	3,000/ kg	Weighted average price observed in Bandung Household Survey
Rice	6,000 /kg	Average price observed in Bandung Household Survey
Fruits	2,500/ kg	Weighted average price observed in Bandung Household Survey
Timber	866,820 / m3	Irawanti et al (2017)
Labour (wages)	60,000 /labour day	Median wage observed in Bandung Household Survey
Seedling costs (all trees)	2,000 / seedling	Average price for coffee seedlings observed in Bandung Household Survey
Seedling costs (fruit trees)	20,000 / seedling	Assumes a semi-mature planting of fruit trees – validated in November 2022 by the project team.
Seedling costs (avocado trees)	50,000 / seedling	Assumes a semi-mature planting of fruit trees – validated in November 2022 by the project team.

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